Sarah K. Swiston

Washington University — Saint Louis, Missouri
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My biological research focuses on developing mathematical models to investigate how species evolve over time and move through space. I am also interested in biology education research, pedagogy, and teaching.

EDUCATION

Washington University | Saint Louis, MO **2020 – 2026**
Ph.D. Candidate in Ecology & Evolutionary Biology (in progress)

University of Oklahoma | Norman, OK **2016 – 2020**
B.A. in Biology, B.S. in Mathematics, minor in Music, Summa Cum Laude

FELLOWSHIPS & AWARDS

National Science Foundation Graduate Research Fellowship Program **2022 – 2026**

Washington University Center for Teaching and Learning Educational Development Graduate Fellowship **2025**

SSB Publisher's Award for Excellence in Systematic Research

ASN Outstanding Student Poster Award

Morris B. Rettner Scholarship

Carl & Pat Bynum Riggs Scholarship **2018**

National Merit Scholar **2016**

Illinois State Scholar

RESEARCH EXPERIENCE

Ph.D. Candidate | Dr. Michael Landis at Washington University **2021 – 2026**
- Biogeography & phylogenetic comparative methods
- R package & Docker image development
- Biogeography of *Liolaemus* lizards, Australian Sphenomorpine lizards, and Hawaiian plants

Rotation Student | Dr. Jonathan Losos at Washington University **2020 – 2021**
- Global distribution of arboreal lizards
- Herpetology field experience

Research Student & Technician | Dr. Katharine Marske at University of Oklahoma **2019 – 2020**
- Lab work including beetle DNA extraction, PCR, and gel electrophoresis
- Data analysis in Geneious, QGIS, BEAST, R

Research Apprentice | University of Oklahoma Biological Station **2018**
- Ant behavioral studies
- Ant heat tolerance testing
- Entomology field experience

Lab Volunteer | Sam Noble Oklahoma Museum of Natural History: Herpetology **2017 – 2018**

- Herpetology field experience
- Specimen preparation and maintenance
- Tissue sampling
- Lab work including DNA extraction and qPCR

PUBLICATIONS

Swiston, S. K., & Landis, M. J. (2025). Testing relationships between multiple regional features and biogeographic processes of speciation, extinction, and dispersal. *Systematic Biology*, 74(2), 282–300.

PREPRINTS & IN PREP

Lichter-Marck, I., **Swiston, S. K.**, Mendes, F. K., May, M., Neupane, S., Rønsted, N., Baldwin, B., Wagner, W., Wood, K., Zapata, F., Landis, M. J. (2025). The Goldilocks effect drives plant diversification on middle-aged Hawaiian islands. *bioRxiv*.

Swiston, S. K., Kuehne, L., Moore, R., & Landis, M. J. (2025). Assessing the efficacy of computational workshops and participatory live coding in evolutionary biology.

Swiston, S. K., McHugh, S., & Landis, M. J. (2025). EMPIRE: The ellipse model for phylogenetic inference of range evolution.

PRESENTATIONS

CIRTL General Meeting **2025**
IGNITE: The Interdisciplinary Graduate Network for Inquiry, Teaching, & Engagement
Co-Presenters: Nicole Strombom

Evolution Conference
EMPIRE: The ellipse model for phylogenetic inference of range evolution

Midwest Ecology and Evolution Conference
Modeling range evolution using ellipses in continuous space

The CIRTL Network Teaching-as-Research Presentations
Assessing the efficacy of computational workshops and participatory live coding in evolutionary biology

Graduate Research Symposium (Washington University)
Assessing the efficacy of computational workshops and participatory live coding in evolutionary biology

Ecology and Evolution Seminar (Washington University) **2024**
Designing and disseminating phylogenetic models of historical biogeography

Evolution Conference
A model of range evolution using ellipses in continuous space

Society for the Study of Amphibians and Reptiles Conference
Testing relationships between regional features and biogeographic processes of speciation, extinction, and dispersal in *Liolaemus* lizards

Evolution Conference **2023**
Testing relationships between multiple regional features and biogeographic process of speciation, extinction, and dispersal

Living Earth Collaborative Seminar (Washington University)
How the environment shapes diversification and dispersal: A feature-informed phylogenetic method of historical biogeography (+ Lizards!)

International Biogeography Society Conference **2022**
Inferring evolutionary rates in a geographic context: Incorporating region features

Midwest Ecology and Evolution Conference
A phylogenetic model of historical biogeography incorporating regional features

Evolution Conference **2021**
Why are arboreal lizards a tropical phenomenon?

POSTERS

Evolution Conference **2025**
EMPIRE: The ellipse model for phylogenetic inference of range evolution

Graduate Research Symposium (Washington University)
Education in phylogenetic modeling: The pedagogy of tutorials, software, and workshops

Society for the Advancement of Biology Education Research Midwest Conference **2024**
Education in phylogenetic modeling: the pedagogy of tutorials, software, and workshops

Evolution Conference
Education in phylogenetic modeling: The pedagogy of tutorials, software, and workshops

University of Oklahoma Undergraduate Research Day **2017**
Amphibious infectious disease in Oklahoma: A citizen science project

THESES

Honors Thesis | University of Oklahoma **2019**
Phylogeography of New Zealand weevil *Geochus tibialis* (Curculionidae)

PRESS

News Article | Neighbors of Glen Ellyn **2024**
A Science Grad Student's Journey

TEACHING

Guest Lecture | Yale University **2026**
Phylogenetic Comparative Methods: Biogeography & Phylogenetic Comparative Methods

- Included a guest lecture, paper discussion, and coding tutorial

Instructor of Record | Mathematical Modeling in Ecology & Evolution **2026**
Co-Instructors: Dr. Abigail Jager

- 1-unit graduate-level course with 12 students
- Classroom activities included lectures, coding exercises, writing, and discussion of primary literature
- Canvas was used to manage course materials

Guest Lecture | Washington University **2025**
Practical Bioinformatics: Shell Scripts

Workshop Instructor | Phylogenetic Biogeography Workshop **2024 – 2025**
Co-Instructors: Michael Landis, Felipe Zapata, Isaac Lichter-Marck, Fábio Mendes, Michael May, Ixchel González Ramírez
Locations: Evolution Conference (2025), Washington University (2024)

- Co-designed curriculum for 3-day and 1-day intensive workshops
- Co-led 3-day workshop at Washington University in 2024
- Led 1-day workshop at Evolution 2025
- Used a combination of lectures, participatory live coding, and group activities

Guest Lecture | Webster University **2024**
Plant Physiology: The Mathematics of Plant Structures

- Used a combination of lecture and think-pair-share

Peer Tutor | Population Genetics

Assistant Instructor | Graduate Research Fundamentals **2023 – 2025**

- Managed course materials in Canvas
- Co-led discussion sections and group activities
- Graded written materials and presentations

Participant | Skype a Scientist **2023 – Present**

Workshop Facilitator | Data Analysis and Visualization in R for Ecologists **2023**
Workshop Instructor: Kaija Gahm, through Carpentries

- Assisted with participatory live coding

Small-Group Facilitator | Assistant Instructor Orientation **08 2023**

- Led orientation activities for graduate students entering their first teaching experiences as Assistant Instructors

Assistant Instructor | Evolution **2022**
Course Instructor: Dr. Ken Olsen

- Led discussion section about primary literature
- Graded student participation and presentation skills
- Graded student essays

Assistant Instructor | Population Genetics **2021**
Course Instructor: Dr. David Queller

- Gave a pre-recorded lecture using Zoom (flipped classroom)
- Co-led classroom recitation focused on problem sets
- Developed and graded student problem sets and exams
- Facilitated group work for coding assignments
- Managed course materials in Canvas

Teaching Team Member | Young Scientist Program **2020 – 2026**

MENTORSHIP

Research Mentor | Washington University **2025**
Mentee: Matthew Pottinger, Washington University in Saint Louis (Undergraduate)

- Assisted in the development of hypotheses and targets for data collection
- Guided the design of statistical analyses using R
- Resulted in an undergraduate thesis: Phylogenetic tests of adaptive and nonadaptive hypotheses for same-sex sexual behavior in mammals
- Currently working toward a manuscript for publication

Copy Editor & Reviewer | Journal of Emerging Investigators **2022 – 2026**

Research Mentor | Young Scientist Program Summer Focus Program **2022**
Mentee: Benjamin Gondzur, Metro Academic and Classical High School (Senior)

- Designed structured, scaffolded research project using R Markdown
- Activities included readings, coding exercises, and data analysis
- Resulted in a high school thesis and an oral presentation

TEACHING PROFESSIONAL DEVELOPMENT

Educational Development Graduate Fellowship | Center for Teaching and Learning **2025**

- Facilitated graduate student learning communities (EPIC, IGNITE)
- Contributed to workshop development

Member | IGNITE Learning Community **2024 – 2026**
(Interdisciplinary Graduate Network for Inquiry, Teaching, and Engagement)

Member | Washington University Education Research Group **2024 – 2026**

Teaching Citation | Washington University **2023**
Professional Development in Teaching Program

Member | Center for Teaching and Learning Graduate Student Advisory Council **2022 – 2024**

Member | EPIC Learning Community **2020 – 2021**
(Exploring Practices in the Classroom)

Member | Washington University in Saint Louis Future Educators **2020**

Competent in the following tools: Canvas, Zoom, Microsoft Office, Google Workspace

WORKSHOPS ATTENDED

Using Game-Inspired Design to Spark Engagement and Deep Learning **2025**
Washington University
Instructor: Dr. Tori Mondelli, University of Missouri

Sparking Joy in Biology Education | SABER 2025 Conference

Scholarship of Teaching and Learning Seminar **2024**
Washington University Center for Teaching and Learning

Introduction to CODON Learning | SABER Midwest 2024 Conference

Data Management | Washington University Libraries Data Services **2023**
Responsible data management, storage, security, and availability

The Inclusive STEM Teaching Project

Evolutionary Quantitative Genetics | Dr. Stevan J. Arnold & Dr. Joe Felsenstein **2022**
Lecturers: Dr. Patrick Carter, Dr. Adam Jones, Dr. Brian O'Meara, Dr. Samantha Price, Dr. Josef Uyeda

Short Teaching Workshops | Washington University Center for Teaching and Learning
Applying Cognitive Science (2021), Engaging in Equity Pedagogy (2021), Introduction to Pedagogical Scholarship (2021), Developing Effective Summative Assessments (2022), Planning and Organizing a Class Session (2023), Process Oriented Guided Inquiry Learning – POGIL (2024), Introduction to Team-Based Learning – TBL (2025), Crafting Critical Thought: Hands-On Creativity for Classroom Engagement (2025)

PROGRAMMING

PhyloDocker | Docker Image: hub.docker.com/r/sswiston/phylo_docker **2022 – Present**

Competent in the following tools: R, R Devtools, R Shiny, RevBayes, Python, Java, Docker, QGIS, Unix

GRADUATE COURSEWORK

Population Genetics, Genomics, Community Ecology, Stochastic Processes, Data Structures & Algorithms, Genomics, Ethics and Research Science

REVIEWED FOR

Evolution, Journal of Emerging Investigators, Methods in Ecology & Evolution, Molecular Phylogenetics & Evolution, Proceedings of the Royal Society B, Systematic Biology

SOCIETY MEMBERSHIPS

American Society of Naturalists (ASN)

National Association of Biology Teachers (NABT)

Partners in Amphibian and Reptile Conservation (PARC)

Professional and Organizational Development Network in Higher Education (POD)

Society for the Advancement of Biology Education Research (SABER)

Society for the Study of Amphibians and Reptiles (SSAR)

Society of Systematic Biologists (SSB)

Society for the Study of Evolution (SSE)

REFERENCES

Dr. Michael Landis
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Relationship: Ph.D. Advisor

Dr. Felipe Zapata
Associate Professor, Department of Ecology and Evolutionary Biology
University of California Los Angeles
612 Charles E. Young Drive South
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Relationship: Collaborator

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Relationship: Teaching Center Fellowship Supervisor